

D898

Algebra for Secondary Mathematics Teaching

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A town of 20,000 people is growing 8% per year. The mayor has served three terms and plans a new library. Which model gives the population after t years?

- A. $P = 20,000 + 0.08t$
- B. $P = 20,000(0.08)^t$
- C. $P = 20,000(1.08)^t$
- D. $P = 20,000(1.8)^t$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

2. [TRANSFER — reversed stem / different variable]

A car's value is modeled by $V = 30,000(0.85)^t$. What is the yearly percent change?

- A. A 15% increase each year.
- B. An 85% decrease each year.
- C. A 0.85% decrease each year.
- D. A 15% decrease each year.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

3.

On a Friday before a long weekend, a teacher asks the class to solve $3^x = 81$. The lesson uses a new set of graphing calculators. Which statement correctly gives x ?

- A. $x = 81 \div 3 = 27$
- B. $x = \log_3 81 = 4$
- C. $x = \log_{81} 3 = 1/4$
- D. $x = \sqrt[3]{81} \approx 4.33$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

4. [TRANSFER — reversed stem / different variable]

If $\log_2 x = 5$, what is x ?

- A. 32
- B. 2.5
- C. 25
- D. 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

5.

A class graphs $f(x) = (x^2 - 9)/(x - 3)$. Several students used a graphing app on tablets the district bought last spring. One student says the graph has a vertical asymptote at $x = 3$. What is correct?

- A. There is a hole at (3, 6) and no vertical asymptote.
- B. There is a hole at $x = -3$.
- C. There are vertical asymptotes at $x = 3$ and $x = -3$.
- D. There is a vertical asymptote at $x = 3$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which function has a vertical asymptote at $x = 2$ and a hole at $x = -1$?

- A. $f(x) = (x - 2) / ((x + 1)(x - 2))$
- B. $f(x) = (x + 1) / (x - 2)$
- C. $f(x) = ((x + 1)(x - 2)) / (x + 1)$
- D. $f(x) = (x + 1) / ((x + 1)(x - 2))$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A student models the height of a design with $f(x) = 2x^2 - 12x + 5$ on a graphing calculator borrowed from the math lab. The design will be printed in blue. What is the vertex?

- A. (-3, 59)
- B. (3, 5)
- C. (3, -13)
- D. (6, 5)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

The parabola $y = x^2 + bx + 7$ has its axis of symmetry at $x = 4$. What is b?

- A. -4
- B. -8
- C. 8
- D. 7

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

The class had just finished a unit on factoring. A teacher asks for the end behavior of $f(x) = -3x^5 + 2x^4 + 100x$. Which description is correct?

- A. It falls to the left and rises to the right.
- B. It rises on both ends.
- C. It follows $100x$, because that term has the largest coefficient.
- D. It rises to the left and falls to the right.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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10.

A table from a textbook published in 2015 lists $x = 0, 1, 2, 3, 4$ with $y = 3, 5, 9, 15, 23$. Which function family fits?

- A. Quadratic.
- B. It cannot be identified without a graph.
- C. Linear, because x increases by 1 each time.
- D. Exponential, because y grows faster and faster.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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11.

A student who finished the warm-up early solves $x^2 = 3x$ by dividing both sides by x and writes $x = 3$. The class is using a new online homework platform. What should the teacher point out?

- A. The answer should also include $x = -3$.
- B. Dividing by x lost the solution $x = 0$; factoring $x(x - 3) = 0$ gives $x = 0$ or $x = 3$.
- C. Dividing both sides of an equation is never allowed.
- D. The answer is complete; $x = 0$ is trivial and can be ignored.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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12.

A teacher plans a lesson around the question “How many handshakes happen if 20 people each shake hands once?” The classroom has movable desks and a large whiteboard. Which plan follows task-based learning?

- A. Offer the task only as enrichment for early finishers.
- B. Have students work alone the whole time with no class discussion.
- C. Pose the problem, let groups explore with lists, tables, or diagrams, then compare strategies and connect them to $n(n - 1)/2$.
- D. Show $n(n - 1)/2$ first, then assign the handshake problem as practice.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A town of 20,000 people is growing 8% per year. ~~The mayor has served three terms and plans a new library.~~ Which model gives the population after t years?

Target word: exponential model **Decides:** 8% growth gives the factor 1.08.

A	Adjacent category	A linear model adding a tiny constant amount, not a percent.
B	Dropped element	Drops the 1; this would shrink the town to 8% each year.
C	Correct	Each year keeps 100% and adds 8%, a factor of 1.08.
D	Confusion set	Reads 8% as 0.8, which is 80% growth.

2. Answer D Concept 1 · transfer

A car's value is modeled by $V = 30,000(0.85)^t$. What is the yearly percent change?

Target word: rate from factor **Decides:** Factor $0.85 = 1 - 0.15$.

A	Inverted operation	A factor below 1 means decay.
B	Confusion set	85% is what remains, not what is lost.
C	Wrong stem number	Reads the factor as a percent.
D	Correct	The car keeps 85% of its value, so it loses 15%.

3. Answer B Concept 2

~~On a Friday before a long weekend,~~ a teacher asks the class to solve $3^x = 81$. ~~The lesson uses a new set of graphing calculators.~~ Which statement correctly gives x ?

Target word: solve for an exponent **Decides:** x is the exponent on 3 that gives 81.

A	Inverted operation	Divides instead of asking for an exponent.
B	Correct	$3^4 = 81$, so the exponent is 4.
C	Confusion set	Swaps the base and the argument.
D	Adjacent category	A cube root solves $x^3 = 81$, where the base is unknown.

4. Answer A Concept 2 · transfer

If $\log_2 x = 5$, what is x ?

Target word: rewrite a log as an exponential **Decides:** $\log_2 x = 5$ means $2^5 = x$.

A	Correct	$2^5 = 32$.
B	Dropped element	Divides 5 by 2, dropping the exponential meaning entirely.
C	Confusion set	5^2 swaps the base and the exponent.
D	Inverted operation	Multiplies 2×5 instead of raising 2 to the 5th.

5. Answer A Concept 3

A class graphs $f(x) = (x^2 - 9)/(x - 3)$. ~~Several students used a graphing app on tablets the district bought last spring.~~ One student says the graph has a vertical asymptote at $x = 3$. What is correct?

Target word: hole or asymptote at $x = 3$ **Decides:** $x^2 - 9 = (x - 3)(x + 3)$; the $(x - 3)$ cancels.

A	Correct	After canceling, $f(x) = x + 3$ with $x \neq 3$; the missing point is $(3, 6)$.
B	Inverted operation	-3 is an x-intercept, not the canceled factor's zero.
C	Confusion set	Treats the numerator's zeros as asymptotes.
D	Dropped element	Skips the factoring step that shows the factor cancels.

6. Answer D Concept 3 • transfer

Which function has a vertical asymptote at $x = 2$ and a hole at $x = -1$?

Target word: build a function from its features **Decides:** $(x + 1)$ must cancel; $(x - 2)$ must remain in the denominator.

A	Inverted operation	Hole at 2 and asymptote at -1 : the roles are reversed.
B	Dropped element	Has the asymptote but no canceled factor, so no hole.
C	Adjacent category	Simplifies to a line with a hole; nothing is left in the denominator.
D	Correct	$(x + 1)$ cancels (hole at -1); $(x - 2)$ remains (asymptote at 2).

7. Answer C Concept 4

A student models the height of a design with $f(x) = 2x^2 - 12x + 5$ ~~on a graphing calculator borrowed from the math lab. The design will be printed in blue.~~ What is the vertex?

Target word: vertex **Decides:** $x = -(-12)/(2 \cdot 2) = 3$.

A	Inverted operation	Drops the negative in $-b/(2a)$; $f(-3) = 18 + 36 + 5 = 59$.
B	Wrong stem number	Uses the constant c as the vertex's y-value.
C	Correct	$x = 12/4 = 3$; $f(3) = 18 - 36 + 5 = -13$.
D	Dropped element	Uses $-b/a = 6$, dropping the 2 in $2a$.

8. Answer B Concept 4 • transfer

The parabola $y = x^2 + bx + 7$ has its axis of symmetry at $x = 4$. What is b ?

Target word: solve for b **Decides:** $-b/(2 \cdot 1) = 4$.

A	Dropped element	Forgets the 2 in the denominator.
B	Correct	$-b/2 = 4$ gives $b = -8$.
C	Inverted operation	Loses the negative sign in $-b/(2a)$.
D	Wrong stem number	The constant term, which does not move the axis.

9. Answer D Concept 5

~~The class had just finished a unit on factoring.~~ A teacher asks for the end behavior of $f(x) = -3x^5 + 2x^4 + 100x$. Which description is correct?

Target word: end behavior **Decides:** Leading term $-3x^5$: odd degree, negative coefficient.

A	Inverted operation	Ignores the negative leading coefficient.
B	Confusion set	That is even-degree, positive-leading behavior.
C	Adjacent category	For large $ x $, the highest power dominates, not the largest coefficient.
D	Correct	Odd degree means opposite ends; the negative sign makes the right end fall.

10. Answer A Concept 6

~~A table from a textbook published in 2015~~ lists $x = 0, 1, 2, 3, 4$ with $y = 3, 5, 9, 15, 23$. Which function family fits?

Target word: function family **Decides:** First differences 2, 4, 6, 8; second differences all 2.

A	Correct	Constant second differences with equal x -steps signal a quadratic.
B	Absolute language	Differences in a table are enough.
C	Dropped element	Checks the inputs but not the outputs' differences.
D	Confusion set	Ratios $5/3, 9/5, 15/9$ are not constant.

11. Answer B Concept 7

~~A student who finished the warm-up early~~ solves $x^2 = 3x$ by dividing both sides by x and writes $x = 3$. ~~The class is using a new online homework platform.~~ What should the teacher point out?

Target word: the error in the solving step **Decides:** Dividing by x assumes $x \neq 0$.

A	Confusion set	Borrows the $\pm$ from equations like $x^2 = 9$; -3 fails here.
B	Correct	Dividing by an expression that can be zero removes that root.
C	Absolute language	Dividing by a nonzero number is fine; the issue is dividing by something that may be zero.
D	Dropped element	$x = 0$ is a true solution: $0^2 = 3 \cdot 0$.

12. Answer C Concept 8

A teacher plans a lesson around the question "How many handshakes happen if 20 people each shake hands once?" ~~The classroom has movable desks and a large whiteboard.~~ Which plan follows task-based learning?

Target word: sequence of a task-based lesson **Decides:** The task comes before the formula.

A	Adjacent category	Treats the core task as an extension instead of the lesson itself.
B	Dropped element	Leaves out the discussion that connects strategies.
C	Correct	Launch, explore, discuss: the formal idea grows out of students' work.
D	Inverted operation	Reverses the order; the task becomes a routine exercise.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Growth factor vs. growth rate	A change of r percent per period gives the factor $(1 + r)$; a factor $b < 1$ means decay of $(1 - b)$ per period.	Using the rate itself (0.08) as the base, or reading 0.85 as an 85% decrease.	A percent change per year paired with an exponential model.
2	Logarithms undo exponentials	$b^x = y$ is the same statement as $\log_b y = x$; a log answers “what exponent?”	Dividing $(81 \div 3)$ or taking a root when the unknown is an exponent.	The variable sits in the exponent.
3	Holes vs. vertical asymptotes	A factor that cancels from numerator and denominator makes a hole; a zero of the denominator that remains after canceling makes a vertical asymptote.	Calling every zero of the denominator an asymptote.	A rational function whose numerator and denominator share a factor.
4	Vertex and axis of symmetry	For $y = ax^2 + bx + c$, the axis is $x = -b/(2a)$ and the vertex's y -value is $f(-b/(2a))$.	Dropping the 2 in $2a$, or the negative sign.	A quadratic in standard form and “vertex,” “maximum,” or “minimum.”
5	Polynomial end behavior	End behavior is set by the leading term: its degree (even or odd) and the sign of its coefficient.	Letting the term with the largest coefficient decide, or ignoring a negative leading coefficient.	“As $x \rightarrow \infty$ ” or “as $x \rightarrow -\infty$.”
6	Reading a function family from a table	With equal x -steps, constant first differences mean linear, constant second differences mean quadratic, and constant ratios mean exponential.	Calling anything that grows faster and faster exponential without checking ratios.	A table of values with equally spaced inputs.
7	Equivalence while solving	Each step must keep the same solution set; dividing by an expression that could be zero can lose solutions, and squaring can add extraneous ones.	Dividing both sides by x to “simplify.”	A variable appears on both sides as a common factor.
8	Task-based learning: launch, explore, discuss	Pose a rich task before showing a method; students explore with their own strategies, then the class compares and connects them to the formal idea.	Teaching the formula first and then assigning the task as practice.	A lesson plan built around one problem.

Course scope (public description): Algebra for Secondary Mathematics Teaching offers an in-depth exploration of advanced algebraic concepts and instructional methodologies tailored for secondary education. The course emphasizes task-based learning, enabling participants to engage deeply with algebra as an extension of number theory, operations, and abstract reasoning. Students will explore key algebraic ideas, including equivalence, patterns of change through covariation, and the integration of multiple representations—such as tables, graphs, equations, and geometric models. The course also examines the historical evolution of algebra and incorporates diverse cultural perspectives. Core topics include variables, functions (including exponential, logarithmic, polynomial, rational, and quadratic functions), and their applications. This course is designed to enhance educators' ability to teach complex algebraic concepts effectively and adapt instructional strategies to meet diverse student needs.